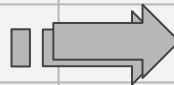


Lab 7

Policy-Based Reinforcement Learning



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A2C, PPO, GAE

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Task 1: A2C on Pendulum
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Task 3: PPO on Walker

Model & Packages

Classes and required packages

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Report + Code + Video

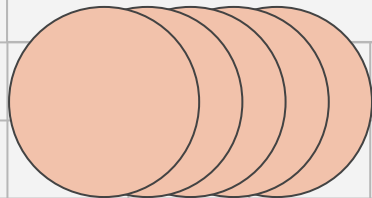
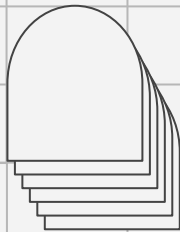
Submission Policy

There will be penalty using the wrong file names!!



Background

Policy-Based RL



A2C: Advantage Actor Critic

- Actor: $\pi_{\theta}(a|s)$

$$J_{\text{actor}}(\theta) = \mathbb{E}_{s \sim d_{\mu}^{\pi_{\theta}}, a \sim \pi_{\theta}(\cdot|s)} [\log \pi_{\theta}(a|s) \cdot A^{\pi_{\theta}}(s, a)] \approx \frac{1}{|B|} \sum_{(s, a, r, s') \in B} \log \pi_{\theta}(a|s) \hat{A}_w(s, a)$$

- Critic: $V_w(s)$

$$L_{\text{critic}}(w) = \mathbb{E}_{(s, a, r, s')} \left[(r + \gamma V_w(s') - V_w(s))^2 \right] \approx \frac{1}{|B|} \sum_{(s, a, r, s')} (r + \gamma V_w(s') - V_w(s))^2$$

- Advantage

$$\hat{A}_w(s, a) = r + \gamma V_w(s') - V_w(s)$$

Algorithm 1 Advantage Actor-Critic (A2C)

```
1: Initialize actor network  $\pi_\theta(a|s)$  and critic network  $V_w(s)$ 
2: for each episode do
3:   Initialize state  $s_0$ 
4:   for  $t = 0$  to  $T$  do
5:     Sample action  $a_t \sim \pi_\theta(a_t|s_t)$ 
6:     Execute  $a_t$ , observe  $r_t, s_{t+1}$ 
7:     Compute TD-error:  $\delta_t = r_t + \gamma V_w(s_{t+1}) - V_w(s_t)$ 
8:     Update critic:  $w \leftarrow w - \alpha_c \nabla_w \delta_t^2$ 
9:     Update actor:  $\theta \leftarrow \theta + \alpha_a \nabla_\theta \log \pi_\theta(a_t|s_t) \cdot \delta_t$ 
10:     $s_t \leftarrow s_{t+1}$ 
11:   end for
12: end for
```

PPO-Clip With GAE

- **Clipped Surrogate Actor Objective**

$$L_{\text{clip}}(\theta) = \mathbb{E}_{(s,a,r,s')} \left[\min \left(\rho_{s,a}(\theta) A^{\text{GAE}}(s,a), \text{clip}(\rho_{s,a}(\theta), 1 - \epsilon, 1 + \epsilon) A^{\text{GAE}}(s,a) \right) \right],$$
$$\approx \frac{1}{|B|} \sum_{(s,a,r,s') \in B} \min \left(\rho_{s,a}(\theta) A^{\text{GAE}}(s,a), \text{clip}(\rho_{s,a}(\theta), 1 - \epsilon, 1 + \epsilon) A^{\text{GAE}}(s,a) \right)$$

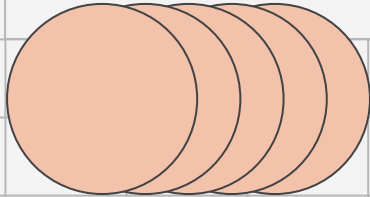
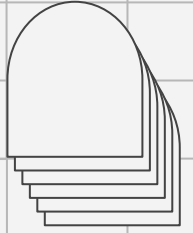
- **Critic Loss:**

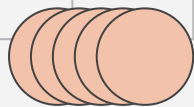
$$L_{\text{critic}}(w) = \mathbb{E}_{(s,a,r,s')} \left[(r + \gamma V_w(s') - V_w(s))^2 \right] \approx \frac{1}{|B|} \sum_{(s,a,r,s')} (r + \gamma V_w(s') - V_w(s))^2$$

- **Overall Objective**

$$J_{\text{PPO}} = J_{\text{clip}}(\theta) - c_1 L_{\text{critic}}(\phi) + c_2 \mathbb{E}_{s \sim d_{\mu}^{\pi_{\theta}}} [H(\pi_{\theta}(\cdot|s))]$$

Lab Description

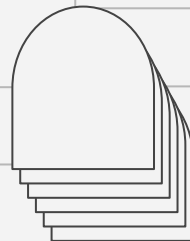




Environment: Pendulum

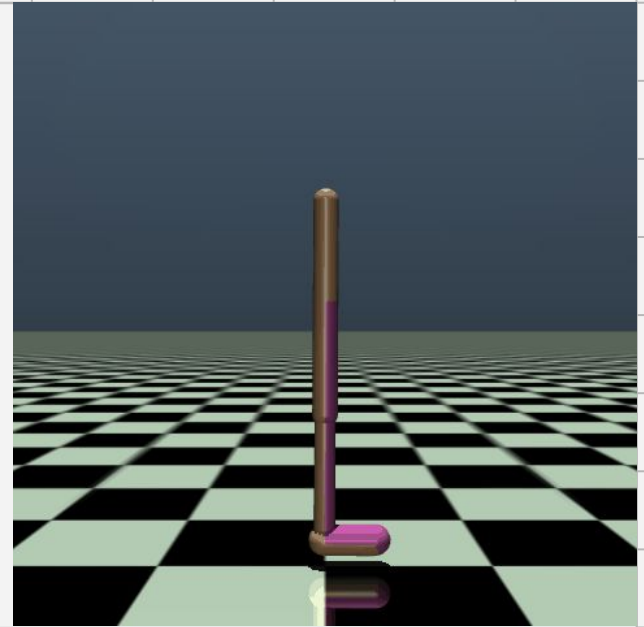
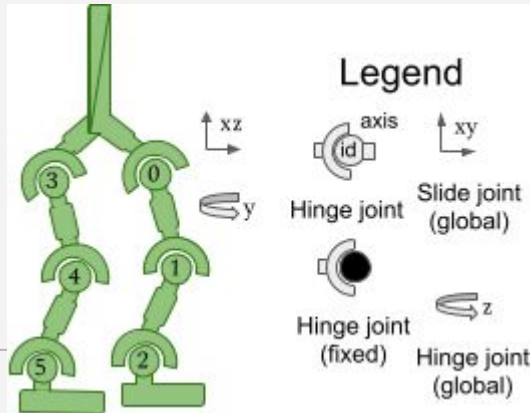


- Goal: swing it into an upright position
- State: $\theta \in [-2, 2], \omega \in [-1, 1]$
- Action: $\tau \in [-2, 2]$
- Observation Space: $[x, y, \omega]$
- Reward:
$$R = -(\theta + 0.1 \frac{d\theta}{dt} + 0.001 \tau^2)$$



Environment: Walker2d-v5

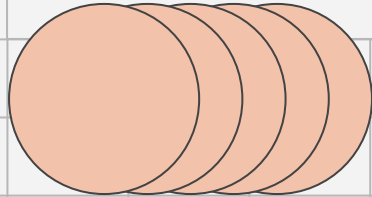
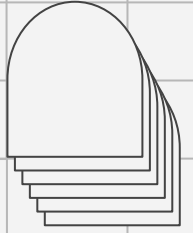
- Goal: Walk in the forward direction
- [Observation Space](#)
- reward = healthy_reward bonus + forward_reward - ctrl_cost
- Action Space:
 - 6 joint torques



Tasks

- Task1: A2C on Pendulum
- Task2: PPO with GAE on Pendulum
- Task3: PPO on Walker2d-v5

Grading Policy



Report

- Introduction (5%): Please provide a high-level introduction to your report. You can mention the most important findings and the overall organization of this report.
- Your implementation (20%): Please briefly explain your implementation for Tasks 1-3. Specifically, please describe:
 - How do you obtain stochastic policy gradient and TD error for A2C?
 - How do you implement the clipped objective in PPO?
 - How do you obtain the estimator of GAE?
 - How do you collect samples from the environment?
 - How do you enforce exploration (despite that both A2C and PPO are on-policy RL methods)?
 - Explain how you use Weight & Bias to track model performance and the loss values (including actor loss, critic loss, and the entropy).

Report

- Analysis and discussions (25%)
 - Plot the training curves (evaluation score versus environment steps) for Task 1, Task 2, and Task 3 separately
 - Compare the sample efficiency and training stability of A2C and PPO.
 - Perform an empirical study on the key parameters, such as clipping parameter and entropy coefficient
 - Additional analysis on other training strategies (Bonus)

Demo Video

- Total Duration: 5–6 minutes
- Language: English (unless pre-approved by TAs)
 - ◆ Source Code (~2 minutes): Describe your implementation
 - ◆ Model Performance (~3 minutes): Demonstrate your obtained models



Model snapshots will NOT be graded if no valid demo video is provided.

Model Snapshots (Up to 50%)

- Task 1

$$\text{Score percentage} = \left(1 - \frac{\max\{0, X - 200k\}}{800k}\right) \times 15\%$$

- Task 2

$$\text{Score percentage} = \left(1 - \frac{\max\{0, X - 200k\}}{800k}\right) \times 20\%$$

- Task 3

Environment steps needed (Reaching score 2500 on Walker2d)	1M	1.5M	2M	2.5M	3M	>3M
Score Percentage	20%	16%	12%	10%	8%	5%

Evaluation Screenshots

You Must

- Include screenshots of your *evaluation results* in your report, a 3-point penalty will be applied for *each* missing screenshot.

```
C:\Users\Red_OAO\Desktop>python test_model_task3.py --model_path LAB5_StudentID_task3_2000000.pt
Environment steps: 20000000, seed: 0, eval reward: 18
Environment steps: 20000000, seed: 1, eval reward: 3
Environment steps: 20000000, seed: 2, eval reward: 5
Environment steps: 20000000, seed: 3, eval reward: 14
Environment steps: 20000000, seed: 4, eval reward: 9
Environment steps: 20000000, seed: 5, eval reward: 12
Environment steps: 20000000, seed: 6, eval reward: 3
Environment steps: 20000000, seed: 7, eval reward: 9
Environment steps: 20000000, seed: 8, eval reward: 5
Environment steps: 20000000, seed: 9, eval reward: 18
Environment steps: 20000000, seed: 10, eval reward: 7
Environment steps: 20000000, seed: 11, eval reward: 18
Environment steps: 20000000, seed: 12, eval reward: 13
Environment steps: 20000000, seed: 13, eval reward: 21
Environment steps: 20000000, seed: 14, eval reward: 4
Environment steps: 20000000, seed: 15, eval reward: 2
Environment steps: 20000000, seed: 16, eval reward: 21
Environment steps: 20000000, seed: 17, eval reward: 5
Environment steps: 20000000, seed: 18, eval reward: 21
Environment steps: 20000000, seed: 19, eval reward: 13
Average reward: 11.05
```

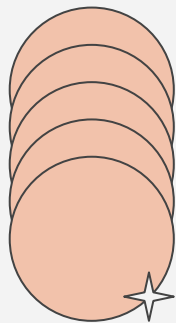
*This is just an example of
an evaluation screenshot*

- Include *commands* to reproduce your results in your report
- Ensure the results in your report are reproducible!!

Submission Policy

Directory Structure

```
LAB7_StudentID.zip
|-- LAB7_StudentID_Code/          <- Source code folder
|   |-- ppo_walker.py
|   |-- ppo_pendulum.py
|   |-- a2c_pendulum.py
|   |-- (any other .py files)    <- Your code files
|   |-- requirements.txt
|   |-- (any other .sh files)    <- Optional
|-- LAB7_StudentID.pdf           <- Technical report (single PDF)
|-- LAB7_StudentID.mp4           <- Demo video (5 - 6 minutes)
|-- LAB7_StudentID_task1_a2c_pendulum.pt    <- Task 1 model snapshot
|-- LAB7_StudentID_task2_ppo_pendulum.pt    <- Task 2 model snapshot
|-- LAB7_StudentID_task3_ppo_1m.pt          <- Task 3 snapshot (step = 1M)
|-- LAB7_StudentID_task3_ppo_1p5m.pt        <- Task 3 snapshot (step = 1.5M)
|-- ...
|-- LAB7_StudentID_task3_ppo_3m.pt          <- Task 3 snapshot (step = 3M)
|-- LAB5_StudentID_task3_best.pt            <- Task 3 snapshot (any step reach score 2500)
```



Thanks for Your Attention

